

Erdős Problem #813

EP-813

L1 - TRACTABLE

PARTIALLY SOLVED

Graph Theory

ORIGINAL PROBLEM STATEMENT

Let $h(n)$ be minimal such that every graph on n vertices where every set of 7 vertices contains a triangle (a copy of K_3) must contain a clique on at least $h(n)$ vertices. Estimate $h(n)$ - in particular, do there exist constants $c_1, c_2 > 0$ such that $n^{1/3+c_1} \ll h(n) \ll n^{1/2-c_2}$?

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>